

Phase 2.2: Proposed Solution

Centralized RESTful Backend Engineering, Architectural Delivery, and AI Capabilities

DOCUMENT REF	PHASE CATEGORY	SYSTEM PATTERN	INTEGRATION MODE
FIT-DOC-PH2.2-SOL	2. Project Design Phase	Model-View-Controller (MVC)	REST API / JSON Standardization

1. SOLUTION OVERVIEW

FITTRACK AI SOLUTION STATEMENT

The FitTrack AI backend provides a secure, scalable, and AI-powered RESTful application specifically engineered to eliminate manual fitness tracking and unguided routines. Developed using Node.js, Express.js, MongoDB, and Mongoose, the backend exposes REST APIs that enable secure user authentication, workout record management, dynamic workout searching, and cognitive exercise intelligence powered by Google Gemini AI.

2. CORE FUNCTIONAL PILLARS OF THE PROPOSED SOLUTION

SECURITY 1. Secure Identity & Auth User registration and login using encrypted password storage via bcrypt.js and stateless JSON Web Tokens (JWT) verified through middleware.	CRUD 2. Centralized Workout Management Comprehensive CRUD operations allowing users to add, view, update, and delete workout logs linked to their authenticated user identifier.
QUERY 3. Multi-Criteria Workout Search Efficient API endpoints that allow users to search and filter previous workout records by workout name, workout category, or workout date.	COGNITIVE 4. Google Gemini AI Guidance Dedicated service layer communicating with Google Gemini AI to provide personalized workout recommendations and performance insights.

3. DOCUMENTED SYSTEM MODULES

The backend separates concerns into modular components ensuring reliability, security, and loose coupling across operations:

MODULE NAME	FUNCTIONAL RESPONSIBILITY	OPERATIONAL MECHANISM
Server Configuration	Central HTTP gateway, JSON body parsing, CORS enablement.	Express.js server initialization (server.js, app.js).
Database Configuration	Connection management and connection lifecycle logging.	Centralized connection via config/db.js with Mongoose.
Authentication	Credential validation, password hashing, JWT generation.	authController.js, bcryptjs, jsonwebtoken.
Workout Management	Full lifecycle CRUD operations on user workouts.	workoutController.js querying models/Workout.js.
AI Recommendation	Dynamic routine generation and statistical insight analysis.	aiController.js interfacing with geminiService.js.
Middleware Layer	Token verification and authorization enforcement.	middleware/auth.js securing protected endpoints.
Error Handling	Centralized JSON error payloads across failure states.	middleware/errorHandler.js intercepting exceptions.

4. SCOPE BOUNDARIES & FUTURE ROADMAP

Source Boundary Rule: In strict alignment with the source document, the active proposed solution implements User Authentication, Workout Management, Workout Search, AI Recommendations, and AI Fitness Insights. The following features are strictly classified as
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5. AI INTEGRATION SPECIFICATIONS

The AI module communicates directly with Google Gemini AI through a dedicated service layer (`geminiService.js`) to evaluate structured user inputs and generate intelligent outputs:

AI CAPABILITY	USER INPUT ATTRIBUTES	GENERATED GEMINI AI DELIVERABLE
AI Workout Recommendation POST <code>/api/ai/workout-recommendation</code>	• Age: User age (Number) • Fitness Goal: Target goal (e.g., Weight Loss) • Experience: Level (Beginner / Intermediate / Advanced)	• Structured personalized workout plan • Weekly exercise schedule and suggestions • Specific exercise sets, reps, and durations • Motivational guidance and safety tips
AI Fitness Insights POST <code>/api/ai/fitness-insights</code>	• Total Workouts: Cumulative completed sessions • Average Duration: Average session minutes • Total Calories Burned: Cumulative calories	• Comprehensive performance summary • Routine consistency analysis • Actionable progressive overload suggestions • Motivational reinforcement

6. SOLUTION BENEFITS & EXPECTED OUTCOMES

OUTCOME DIMENSION	TRADITIONAL MANUAL TRACKING	FITTRACK AI PROPOSED SOLUTION
Data Integrity	Scattered paper notes, vulnerable to errors and loss.	Centralized MongoDB storage enforced by Mongoose validation.
Access Control	No protection for personal health records.	Cryptographic hashing (bcrypt) and stateless JWT verification.
Record Accessibility	Time-consuming manual page flipping or filtering.	Rapid search indexing across name, category, and date parameters.
Fitness Guidance	Static, unguided exercise selections.	Intelligent Google Gemini AI recommendations tailored to goals.
System Interoperability	Isolated, non-digitized records.	Standardized JSON responses ready for Web, Mobile, and Postman.

7. VERIFICATION & OPERATIONAL DELIVERY

The proposed backend solution is fully verified through API test execution (using Postman and ThunderClient) covering the complete operational cycle:

STAGE	API EXECUTION SET	VERIFICATION CONFIRMATION
Step 1	Authentication APIs	Verified registration (HTTP 201), login (HTTP 200), and profile retrieval with JWT validation.
Step 2	Workout Management APIs	Verified workout creation, listing, ID query, updating, and deletion with ownership checks.
Step 3	AI Integration APIs	Verified live Google Gemini communication for routine plans and performance analytics.